Write your questions and thoughts here!

Logarithmic Function Manipulation Graphically

Expand the function: $f(x) = \log_2(4x)$

What happens to the graph?

Expand the function: $f(x) = \ln(x-4)^2$

What happens to the graph?

Try on your own!

Find the domain/range and describe any transformations on each function.

$\log_3(18 - 9x)$

2.12 Logarithmic Function Manipulation

AP Precalculus

2.12 Practice

Let x and y be positive constants. Write each expression as a sum or difference of logarithms.

1. $\log_3(xy^3)$

2. $\ln\frac{x^2}{y^3}$

3. $\log(1000x^3)$

4. $\log_2\left(8\sqrt{x}\right)$

5. $\log_2\left(\sqrt{\frac{16x^3}{y^2}}\right)$

6. $\log_3(9x - 27)$

Let x and y be positive constants. Write each as a single logarithm.

7. $\ln 3 + \ln x$

8. $2\log_3 x - 4\log_3 y$

9. $\frac{1}{2}(\log x + 3\log y)$

10. $3\log_5(x-4) - 2\log_5 y$

CALCULATOR ACTIVE. Use the change of base to change each to a logarithm with base 10 or base e. Then use a calculator and find the value of the logarithm to the nearest thousandth.

11. $\log_4 123$

12. $\log_9 578$

Describe any transformations on the function, then find the domain and range of the transformed function.

13. $f(x) = \log_4(16x)$

14. $g(x) = \log_2\left(\frac{x}{8}\right)$

15. $f(x) = \log_5 x^2$

16. $h(x) = \log_2(32 - 16x)$

17. Let x and y be positive constants. Which of the following is equivalent to $5\log x + 6\log y$.

(A) $\log(x^5 + y^6)$ (B) $\log(x^5 y^6)$ (C) $\log(x^5 - y^6)$ (D) $\log(5x + 6y)$

---

18. Let x and y be positive constants. Which of the following is **NOT** equivalent to
$$\log_3(x+3) + 2\log_3(x-1) - \log_3(x+2)$$

(A) $\log_3\left(\frac{2(x+3)(x-1)}{(x+2)}\right)$

(B) $\log_3\left(\frac{(x+3)(x-1)(x-1)}{(x+2)}\right)$

(C) $\log_3\left(\frac{(x+3)(x-1)^2}{(x+2)}\right)$

(D) $\log_3\left(\frac{(x+3)(x^2-2x+1)}{(x+2)}\right)$

---

19. If $\log 3 = x$ and $\log 4 = y$, which of the following expresses $\log\sqrt{12}$?

(A) $\frac{1}{2}(x-y)$ (B) $\frac{1}{2}(xy)$ (C) $\frac{1}{2}x + 2$ (D) $\frac{1}{2}(x+y)$

---

20. Approximate $\log_b 18$, given that $\log_b 2 \approx 0.4307$ and $\log_b 3 \approx 0.6827$.

(A) 0.8968 (B) 1.1134 (C) 1.5441 (D) 1.7961

2.12 Logarithmic Function Manipulation

AP Precalculus Name: _____ **CA #1**

Instructions: Let x and y be positive constants. Write each as a single logarithm.

1. $2log_4 x - 6log_4 y$

2. $6\ln x + \frac{1}{2}\ln y$

3. $\frac{1}{3}(\log x + 2\log y)$

4. $\frac{1}{2}log_5 x^3 - 2log_5 y$

Instructions: Let x and y be positive constants. Write each expression as a sum or difference of logarithms.

5. $\ln x^4 y$

6. $log_3\sqrt{\frac{x^2}{y}}$

7. $log_6\sqrt{x^3 y^5}$

8. $\log(10x - 30)$

Instructions: Find the domain/range and describe any transformations on the function.

9. $log_2(8x)$

Domain: Transformations:

Range:

10. $log_3(27 - 27x)$

Domain: Transformations:

Range:

Answers

1. $log_4 \frac{x^2}{y^6}$
2. $\ln x^6\sqrt{y}$
3. $\log \sqrt[3]{xy^2}$
4. $log_5 \frac{\sqrt{x^3}}{y^2}$
5. $4\ln x + \ln y$
6. $\frac{1}{2}(2log_3 x - log_3 y)$
7. $\frac{1}{2}(3log_6 x + 5log_6 y)$
8. $1 + \log(x - 3)$
9. Domain: $(0, \infty)$ Range: $(-\infty, \infty)$ Transformations: Vertical translation up 3.
10. Domain: $(-\infty, 1)$ Range: $(-\infty, \infty)$ Transformations: Vertical translation up 3, horizontal translation right 1. Horizontal reflection.

AP Precalc

Write your questions and thoughts here!

2.13A Exponential and Logarithmic Equations and Inequalities

2.13A Notes

EXAMPLE ONE: $\log_3 x + \log_3(x+7) = \log_3 60$

IF
$$\boldsymbol{log_a B = log_a C}$$

THEN
$$\boldsymbol{B = C}$$

EXAMPLE TWO: $3 - \log_2(x+7) = \log_2 5$

Solve an equation with logs and constants.

EXAMPLE THREE: $3(2)^{x+2} = 12$

Use logs to solve exponentials!

EXAMPLE FOUR: $\log(x+6) + 3 = 2$

Use technology!!!

Write your questions and thoughts here!

Up for the challenge? Try these!

Ex 5: $\ln\sqrt{x-12}=4$

Ex 6: $9\left(\frac{1}{27}\right)^{\frac{x}{3}}=81$

Ex 7: $\log(2x+7)+\log 3=\log(x+6)$

2.13A Exponential and Logarithmic Equations and Inequalities

AP Precalculus

2.13A Practice

CALCULATOR ACTIVE. Solve each equation. Have both the exact answer and the answer rounded to three decimal places.

1. $\log_5 x=2$

2. $e^{-x}=3.65$

3. $\log x+\log(x-3)=1$

4. $80e^{0.045x}=240$

5. $\log_3(5-2x) = \log_3(3x+1)$

6. $3 - \log_4(x+3) = 5$

7. $\ln 12 = \ln(2x+3) - \ln(x-4)$

8. $e^{2x-1} + 68 = 207$

9. $\log_2(3x-52) - 4 = \log_2 x$

10. $\log_3 \sqrt{2x+3} = 1$

11. $\ln x + \ln(x - 10) = \ln 24$

12. $3(2^{x+4}) - 12 = -10$

CALCULATOR ACTIVE. Solve each equation with a graphing calculator. Round to three decimal places.

13. $0.5e^{\frac{x}{4}} = 12$

14. $\frac{3}{4}\ln\left(\frac{4}{5}x\right) + 10 = 12$

15. Use the formula for continuously compounded to solve. $A = Pe^{rt}$, where A is how much money we currently have, P is the principal (how much we started with), r is the interest rate and t, is the amount of time in years.

    Mr. Kelly currently has $450,000 in an investment account. He originally put in $200,000 into the account which earns 4.5% interest. How many years has he been investing in this account?

16. Consider the functions f and g given by $f(x) = \ln(-x + 8)$ and $g(x) = \ln(x + 2) + \ln(x - 8)$. In the xy-plane, what are all the x-coordinates of the points of intersection of the graphs of f and g?

(A) $x = 8$

(B) $x = -3$

(C) $x = 8$ and $x = -3$

(D) No Solution

---

17. Solve the equation $\log_b a + \log_b 5 = c$ for a.

(A) $\frac{5}{b^c}$

(B) $5b^c$

(C) $b^c - 5$

(D) $\frac{b^c}{5}$

---

18. What are all values of x for which $\ln(x^5) = \ln(x^3) + 16$?

(A) $x = e^8$ and $x = -e^8$

(B) $x = -e^8$

(C) $x = e^8$

(D) $x = 8$ and $x = -8$

2.13A Exponential and Logarithmic Equations and Inequalities

AP Precalculus Name: _____ **CA #1**

CALCULATOR ACTIVE: Instructions: Solve each equation. Have exact answer and answer rounded to nearest thousandth.

1) $log_2(x-3)=5$

2) $2(3^{4x})=40$

3) $\log(x-3)+\log(x-4)=\log(7-x)$

4) $8e^{10x}=640$

5) $log_6(3)+log_6(x-8)=log_6(4x-26)$

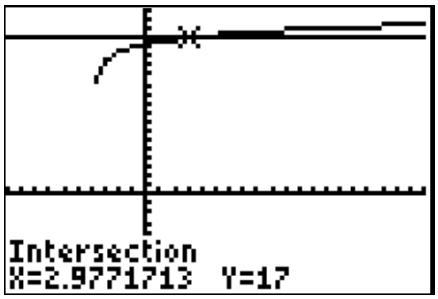
6) $15+log_2(2x)=20$

CALCULATOR ACTIVE: Instructions: Solve each equation with a graphing calculator. Round to nearest thousandth.

7) $\frac{4}{5}log_2\left(2x+\frac{15}{2}\right)+14=17$

ANSWERS

1) 35
2) $\frac{log_3 20}{4} \approx 0.682$
3) 5
4) $\frac{\ln 80}{10} \approx 0.438$
5) No Solution
6) 16
7) 2.977



AP Precalc

Write your questions and thoughts here!

2.13B Exponential and Logarithmic Equations and Inequalities

2.13B Notes

Solve the following inequalities.

Ex 1: $\log(x-8)+\log 2 \le \log(3x+4)$

First…what restrictions are there with the arguments of these logs?

So…..

Now solve and compare the intersections!

Try this one!

Ex 2: $\log_2(2x+6) > 3$

*When changing from logarithmic to exponential the argument stays on the side it started!

*What would happen if this was less than instead?

Do exponentials have restrictions on their domains?

Solve this one!

Ex 3: $3^{x+2}-4 > 23$

Find the inverse of each function.

Ex 4: $f(x) = 3(2^{x+1})-4$

Ex 5: $g(x) = \ln(2x+3)+10$

2.13B Exponential and Logarithmic Equations and Inequalities

AP Precalculus

2.13B Practice

CALCULATOR ACTIVE. Solve each inequality.

1. $\log_4 x < 3$

2. $2^{x+3} + 5 > 37$

3. $\log 5 + \log(x - 2) \geq \log(3x + 8)$

4. $4(3^{2x}) - 8 \leq 316$

5. $\log_2(x + 5) - 8 > -3$

6. $\ln(x + 4) < \ln(x - 6) + \ln 3$

Find the inverse of each function.

7. $g(x) = \ln(x - 4) + 8$

8. $f(x) = 3(2^x) + 6$

9. $h(x) = 3\log_2(2x + 1) - 3$

10. $j(x) = 2e^{x+8} - 5$

11. Use the formula for continuously compounded to solve. $A = Pe^{rt}$, where A is how much money we currently have, P is the principal (how much we started with), r is the interest rate and t, is the amount of time in years.

    If Mr. Brust currently has $250,000 in his retirement account that earns him 8.5% annual interest, how long will it take for the account to have at least $1,000,000 if he does not add any more money into the account?

12. When considering the equation $\log(x-3)+\log(5)>\log(x+9)$, which of the following domains is our initial restriction.

(A) $(3,\infty)$

(B) $(5,\infty)$

(C) $(-9,\infty)$

(D) $(6,\infty)$

---

13. When considering the equation $\log(x-3)+\log(5)>\log(x+9)$, which of the following represents the domain of all solutions to the inequality?

(A) $(3,\infty)$

(B) $(5,\infty)$

(C) $(-9,\infty)$

(D) $(6,\infty)$

---

14. Express y as a function of x. A, B and C are constant, positive numbers.

$$\log(y-A)=Bx-\log C$$

(A) $y=\frac{10^{Bx}}{C}+A$

(B) $y=C+A(10^{Bx})$

(C) $y=\frac{C}{A}-10^{Bx}$

(D) $y=\frac{Bx^{10}}{C}+A$

2.13B Exponential and Logarithmic Equations and Inequalities

AP Precalculus Name: _____

CA #1

CALCULATOR ACTIVE: Instructions: Solve each inequality.

1) $log_5(x+8)-6<-4$

2) $8\cdot 4^{2x}-5>-3$

3) $\log(3x-2)\geq \log 5+\log(x-4)$

4) $log_3\,(2x-3)\geq 3$

Instructions: Find the inverse of each function.

5) $g(x)=\log(2x-3)-5$

6) $f(x)=2(3^{x+1})+10$

1) $-8 < x < 17$ or $(-8,17)$
2) $x > -\frac{1}{2}$
3) $4 < x \leq 9$ or $(4,9]$
4) $x \geq 15$ or $[15, \infty)$
5) $g^{-1}(x) = \frac{10^{x+5}+3}{2}$
6) $f^{-1}(x) = log_3\left(\frac{x-10}{2}\right) - 1$

AP Precalc

Write your questions and thoughts here!

2.14 Logarithmic Function Context and Data Modeling

2.14 Notes

QUICK REVIEW

Remember that logarithmic functions have independent values that increase proportionally while the dependent values increase at a constant rate.

Independent - x	1	3	9	27	81
Dependent - $f(x)$	3	5	7	9	11

*The amount the proportional values increase of the independent values would also be the base.

Ex 1: At the end of the school year, the classroom gets really warm! As the number of students increases so does the temperature. Mr. Bean took some data last May!

# of Students	2	4	8	16
Temperature	67	69	71	73

a. What's a regression equation to model the situation?

b. According to the model, what is the temperature if there are 35 students?

c. According to the model, how many students are there when the temperature reaches 85 degrees?

Ex 2: Mr. Kelly loves to wake up and have a fresh cup of coffee. He likes his coffee to cool a bit before drinking it. He makes a table of values to help him find an appropriate model. He sets his data up so that the time (in minutes) is a function of the temperature (in degrees).

Temp (T)	95	90.6	86.5	82	78	75
Time (M)	0	3	6	9	12	15

a. Find a regression equation to model the equation:

b. How long will it take for Mr. Kelly's morning coffee to be 70 degrees?

Write your questions and thoughts here!

Ex 3: People measure loudness in terms of decibels using the following function, where $D(I)$ is the decibel level, I is the intensity of the sound and 10^{-12} is the intensity of the quietest sound a human can hear.

$$D(I) = 10\ log\left(\frac{I}{10^{-12}}\right)$$

What's the intensity of the sound of the school fire alarm if it registers as 100 decibels?

Ex 4: Mr. Brust wants to save for a new Lamborghini. He currently has $50,000 in an account that pays interest continuously and will need $300,000 in total. If he leaves his money in the account for 12 years and does not add anything, what interest rate would he need to achieve his goal?

Use the equation below where $A(t)$ is the current amount of money, P is the principal (starting value), r is the interest rate and t is the time in years.

$$A(t) = Pe^{rt}$$

2.14 Logarithmic Function Context and Data Modeling

AP Precalculus

2.14 Practice

Calculator Active. Use the data provided to find a regression equation and answer the questions.

1. Mr. Bean started learning Chinese in hopes of translating all his videos one year. He created some data to find a model for the number of days studying as a function of the number of words he learned that day.

# of Words Learned	10	32	79	197	400
# of Days Studying	1	6	10	14	17

a. What is a logarithmic regression equation that could model this situation?

b. How many days would it take for Mr. Bean to learn 1,000 words?

c. How many words would he learn if he studied for 2 months (60 days)?

2. Mr. Sullivan recently decided (with the help from his doctor) that he needed to start a diet. Each week he collects data on the amount of weight he loses that week. He wants to find a regression equation that models the pounds lost on a particular week as a function of the week number.

Week #	1	2	6	9
Pounds lost that week	9.5	8.5	6.5	5.5

a. What's a logarithmic regression equation that could model this situation?

b. During which week will Mr. Sullivan lose 2 pounds?

c. How many pounds will Mr. Sullivan lose during the 12th week?

---

3. The latest and greatest new tech firm, Math Geek Tech, is about to go public on the stock exchange. The market guys at Math Geek Tech want to create a regression model that shows the stock price (in dollars) as a function of the number of days on the market.

Days on Market	1	3	4	7
Stock Prices ($)	112	150.5	160.5	180.1

a. What's a logarithmic regression equation that could model this situation?

b. How many days will it take for the stock price to be $250?

c. What would the stock price be after the first two weeks on the market (14 days)?

Calculator Active. Use the information given to answer the questions. Round/truncate to three decimal places.

4. People can use the formula below to determine future population $N(t)$ of cities. N_0 represents the initial population, r is the rate of population growth, and t is the time in years.

$$N(t) = N_0 e^{rt}$$

   a. What is the population of Cleveland, Ohio in 10 years if there are currently 275,000 people, with a population growth rate of 2.5%.

   b. What growth rate would Cleveland, Ohio need to achieve a population of 400,000 people in 20 years?

5. Forensics often use Newton's Law of Cooling to determine the elapsed time since a person has died. The formula is $t = -10\ ln\left(\frac{T-R}{98.6-R}\right)$, where T stands for the body's temperature in degrees Fahrenheit, R is the temperature of the room and t is the elapsed time since death in hours.

   a. How many hours elapsed if the temperature of the room was 75° and the body's temp was 85°?

   b. Suppose a body was found 5 hours after death (neighbor heard thud) in a room that was 65°. What was the temperature of the body when it was found?

2.14 Logarithmic Function Context and Data Modeling

2.14 Test Prep

6. The table presents values for a function f at selected values of x. A logarithmic regression is used to model the data. What is the value of $f(4.5)$?

x	$f(x)$
1	44.9
3	37.9
7	32.4
9	30.7

(A) 35.225 (B) 34.987 (C) 54.639 (D) 54.012

7. **Calculator active.**

x	4	8	16	32	64
$f(x)$	1	5	9	13	17

Let f be an increasing function defined for $x > 0$. The table gives values of $f(x)$ for selected values of x. The function g is given by $g(x) = \frac{x^2-3x+5}{x-3}$.

(A)

i. The function h is defined by $h(x) = (g \circ f)(x) = g\big(f(x)\big)$. Find the value of $h(4)$ as a decimal approximation, or indicate that it is not defined.

ii. Find the value of $f^{-1}(64)$, or indicate that it is not defined.

(B)

i. Find all zeroes of $g(x)$, as decimal approximations, or indicate there are no such values.

ii. Determine the end behavior of g as x decreases without bound. Express your answer using the mathematical notation of a limit.

(C)

i. Use the table of values of $f(x)$ to determine if f is best modeled by a linear, quadratic, exponential, or logarithmic function.

ii. Give a reason for your answer based on the relationship between the change in the output values of f and the change in the input values of f.

2.14 Logarithmic Function Context and Data Modeling

AP Precalculus Name: _____ **CA #1**

Instructions: Use the information given to answer the questions. Round to nearest thousandth.

1) People can use the formula below to determine future populations (*N(t)*) of cities. N_0 represents the initial population, *r* is the rate of population growth, and *t* is the time in years.

$$N(t) = N_0 e^{rt}$$

a) What would the population be of Rochester, NY be in 10 years if there are currently 210,000 people, with a population growth rate of 1.2%.

b) What growth rate would Rochester, NY need to achieve a population of 250,000 people in 30 years?

Instructions: Use the data provided to find a regression equation and answer the questions.

2) The amount bacteria (in thousands) found on the beach has been steadily declining. Scientists want to create a regression model to show the number of bacteria as a function of years.

Year	1	2	3	4
Bacteria (in thousands)	9.9	7.8	6.7	5.8

a) What's a logarithmic regression equation that could model this situation?

b) How many years until there are 2 thousand bacteria in a sample on the beach?

c) How many bacteria can be found in a sample after 6 years?

3) Scientists are studying the revitalization efforts to restore the wolf population in Wyoming. Each year they find how many wolves there are as an average of the previous 12 months.

Year	2	5	6	8
Average Wolves	21.6	28.9	30.3	32.6

a) What's a logarithmic regression equation that could model this situation?

b) How many wolves can we predict there will be in year 10?

c) How many years for there to be 50 wolves on average for the year?

Answers

1) A) 236,774 people
   B) 0.5%

2) A) $f(x) = 9.858 - 2.939 \ln x$
   B) x = 14.6, so 15 years
   C) 4.6 thousand bacteria

3) A) $f(x) = 16.106 + 7.933 \ln x$
   B) 34.372, so 34 wolves
   C) 71.713, so 72 years.

AP Precalc

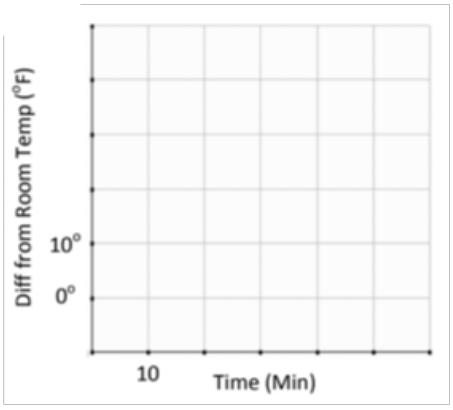
Write your questions and thoughts here!

2.15 Semi-Log Plots

2.15 Notes

When Sully is ready to retire, he has plans on moving to New York City to become a butcher. In fact, he wants to open his own butcher shop, "The New York Metzgerei," where he can sell his signature product: **Sullamy Picante**! Sully has to cook the meat and then let it cool while recording the temperature during the production process. One day, Sully observes the following temperatures:

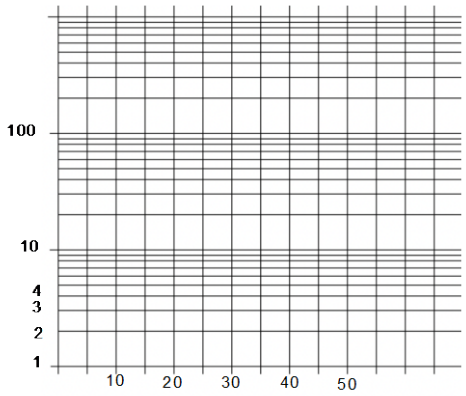
Time (min)	10	14	20	22	26	30	36	40	42	44
Temperature (degrees above room temp in °F)	51	41	30	26	21	17	11	8	6	5



What type of function does this model?

Semi-Log Plot:

Log Scale goes:



What happens to an exponential function on semi-log graph?

Find a regression equation to model the above exponential function:

Take the log and rewrite the equation. What kind of function do you have now?

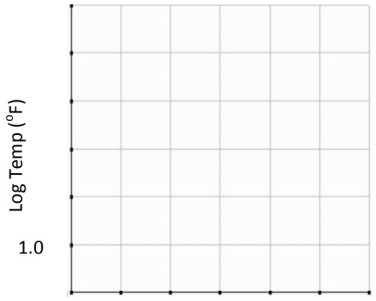
Complete the table:

Time (min)	10	14	20	22	26	30	36	40	42	44
Log (Temp)										

To "straighten" the data, take the common log of each of the temperatures. (Log L_2 → L_3)

Plot the new graph. What do you notice?

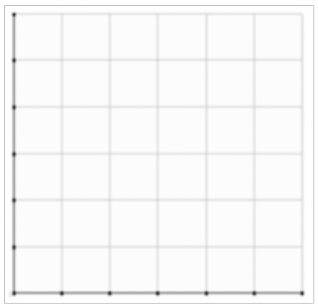
Find a linear regression model of the new data.

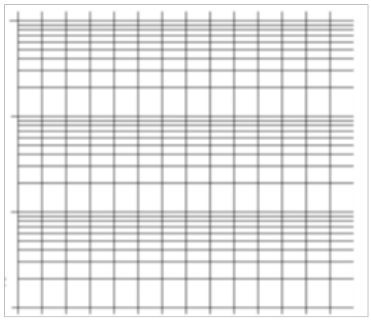


Write your questions and thoughts here!

Ex 1: Plot on the normal graph and on the semi-log graph. Is the data exponential? How do you know? If it is exponential, find a regression equation for the data.

x	2	3	5	7	8
Y	10.1	21.3	93.9	414.3	869.9





x	2	3	5	7	8
Y	10.1	21.3	93.9	414.3	869.9
Log y					

Complete the table to find log y, then find a linear regression model of (x, log y).

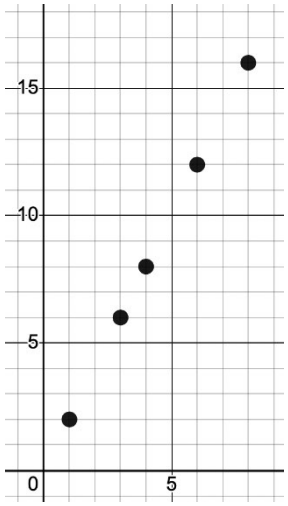
2.15 Semi-Log Plots

AP Precalculus

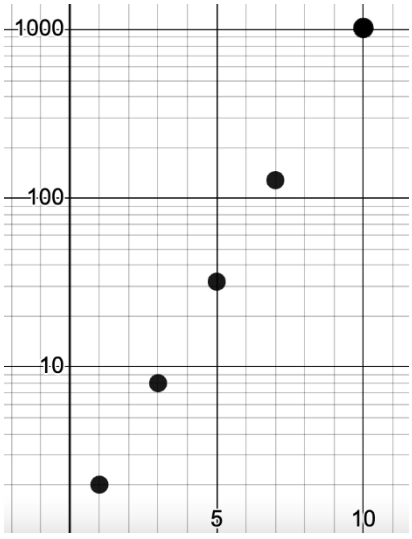
2.15 Practice

Tell which graphs represent exponential functions and which do not. Then explain why.

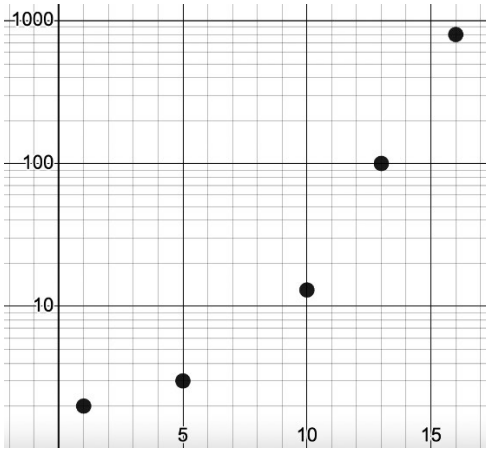
1.



2.



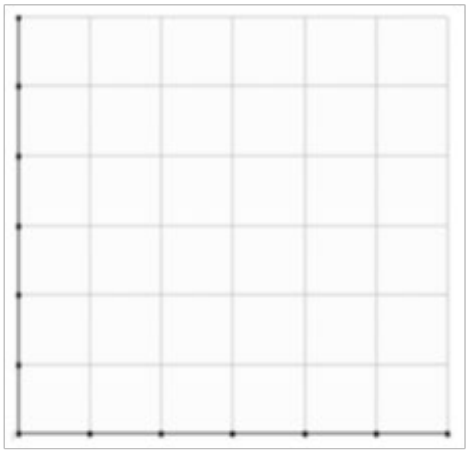
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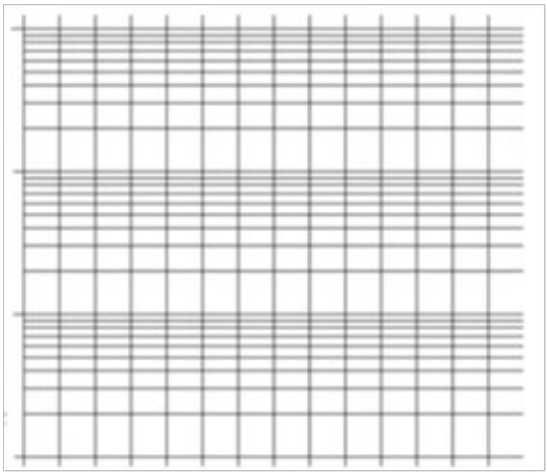


CALCULATOR ACTIVE. Answer the questions pertaining to the given data.

4. a. Plot the following data on both graphs below.

x	2	4	5	8	9
y	7	15	23	77	115





b. Find a regression equation for the above data.

c. Take the log of both sides and use log rules to create a linear function.

d. Complete the table to find $\log y$.

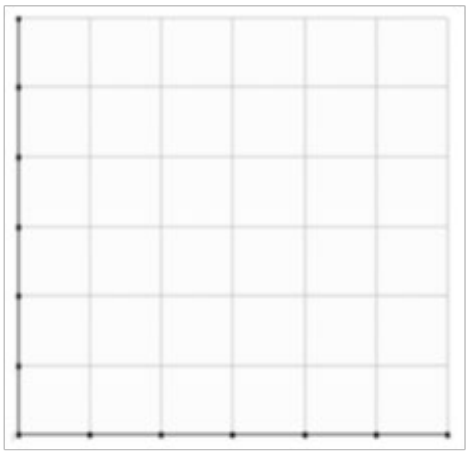
x	2	4	5	8	9
y	7	15	23	77	115
$\log y$					

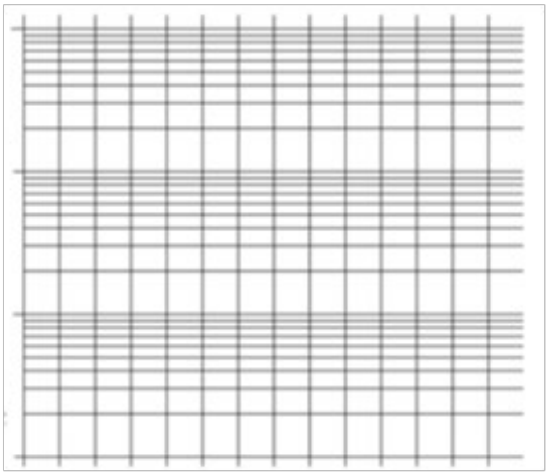
e. Find a linear regression equation for $(x, \log y)$.

---

5. a. Plot the following data on both graphs below.

x	1	3	4	5	7
y	65.0	32.5	16.3	8.2	1.0





b. Find a regression equation for the above data.

c. Take the log of both sides and use log rules to create a linear function.

d. Complete the table to find $\log y$.

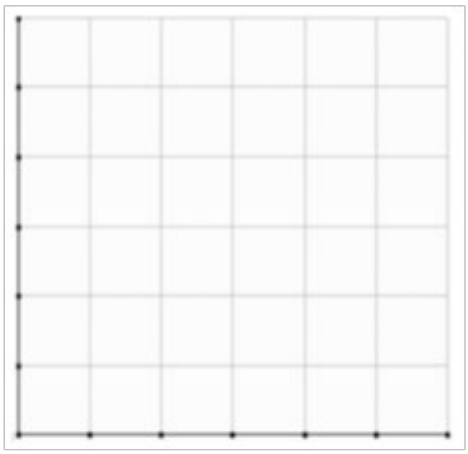
x	1	3	4	5	7
y	65.0	32.5	16.3	8.2	1.0
$\log y$					

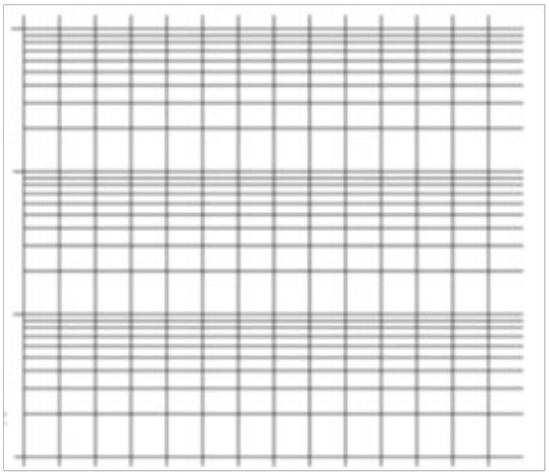
e. Find a linear regression equation for $(x, \log y)$.

---

6. a. Plot the following data on both graphs below.

x	1	3	4	5	7
y	10.5	46.3	97.2	204.2	900.5





b. Find a regression equation for the above data.

c. Take the log of both sides and use log rules to create a linear function.

d. Complete the table to find $\log y$.

x	1	3	4	5	7
y	10.5	46.3	97.2	204.2	900.5
$\log y$					

e. Find a linear regression equation for $(x, \log y)$.

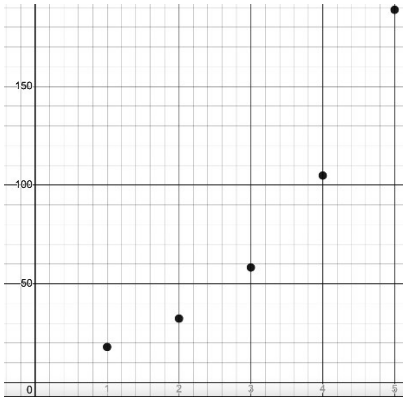
2.15 Semi-Log Plots

2.15 Test Prep

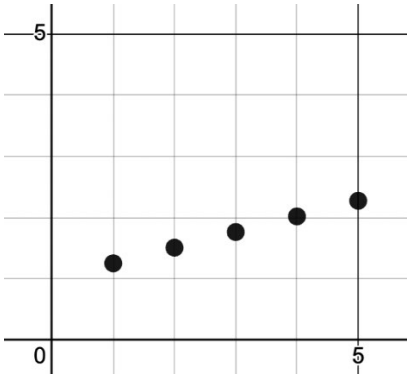
7. The table gives values for a function f at selected values of m. Which of the following graphs could represent these data in a semi-log plot where the vertical axis is logarithmically scaled.

m	1	2	3	4	5
$f(m)$	18	32.4	58.3	104.9	188.9

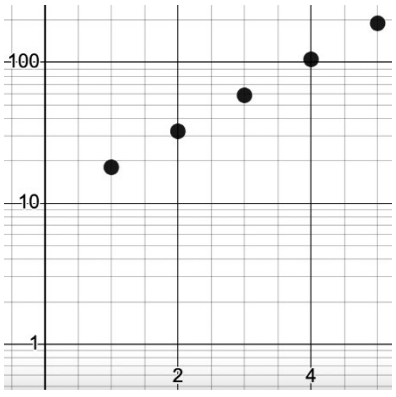
(A)



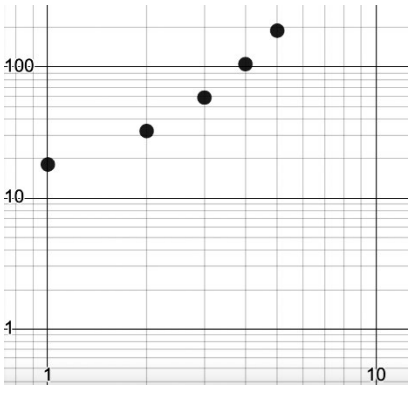
(B)



(C)



(D)



8.

- Unless otherwise specified, the domain of a function *f* is assumed to be the set of all real numbers *x* for which *f(x)* is a real number.
- Solutions to equations must be real numbers. Determine the exact value of any expression that can be obtained without a calculator. For example, $\log_2 8$ can be evaluated without a calculator.
- Unless otherwise specified, combine terms using algebraic methods and rules for exponents and logarithms, where applicable. For example, $2x + 3x$, $5^2 \cdot 5^3$, $\frac{x^5}{x^2}$, and $\ln 3 + \ln 5$ should be written in equivalent forms.
- For each part of the question, show the work that leads to your answers.

(A) The functions g and h are given by: $g(x) = 4\log x + \frac{1}{3}\log x$ and $h(x) = \frac{3^5\sqrt{3^x}}{3^2}$

(i) Rewrite $g(x)$ as a single natural logarithm without negative exponents in any part of the expression.

(ii) Rewrite $h(x)$ as an expression as a single base with an exponent.

(B) The functions j and k are given by: $j(x) = \frac{(e^2)^x}{e^9}$ and $k(x) = \log_2(x + 8) + \log_2 x$

(i) Solve $j(x) = e$ for values of x in the domain of j.

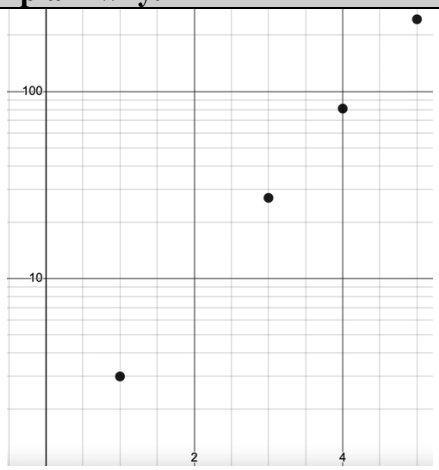
(ii) Solve $k(x) = 5$ for values of x in the domain of k.

2.15 Semi-Log Plots

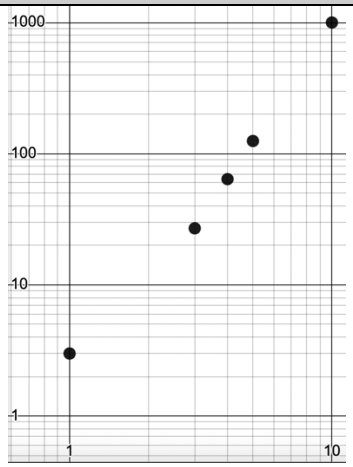
AP Precalculus Name: _____ **CA #1**

Instructions: Instructions: Tell which graphs represent exponential functions and which do not. Then explain why.

1)



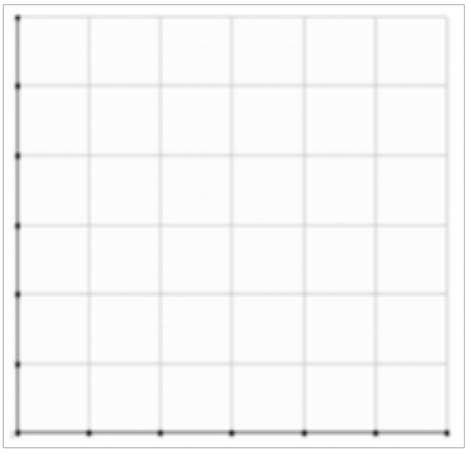
2)

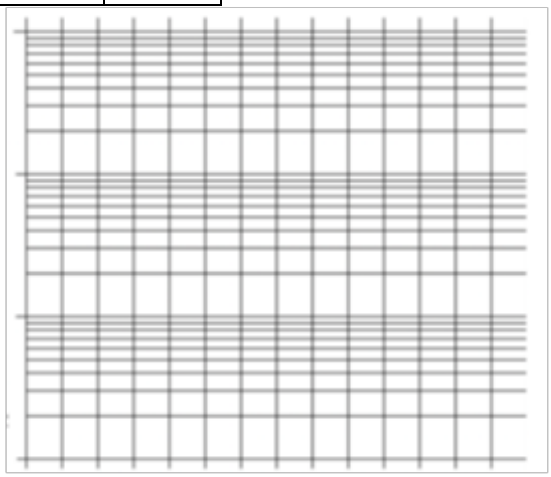


CALCULATOR ACTIVE: Instructions: Answer the questions pertaining to the given data.

3) A) Plot the following data on both graphs below.

X	1	3	4	6	7
Y	3.6	20.74	49.77	286.65	687.97





B) Find a regression equation for the above data.

C) Take the log of both sides and use log rules to create a linear function.

D) Complete the table to find log y.

X	1	3	4	6	7
Y	3.6	20.74	49.77	286.65	687.97
Log y					

E) Find a linear regression equation for (x, log y).

1) Yes , because the semi-log plot shows a linear relationship
2) No because the linear relationship is on a log-log plot.
3) A) graphs should be able to be done, if you need help, please see your teacher.
   B) $f(x) = 1.5(2.4^x)$
   C) $\log y = \log 1.5 + x \cdot \log 2.4$
   D)

X	1	3	4	6	7
Y	3.6	20.74	49.77	286.65	687.97
Log y	0.56	1.32	1.70	2.46	2.84

   E) $\log y = 0.38x + 0.176$

Name: _____ Date:_____ Period: _____ **Review**

End-of-Unit 2B Review – Exponential and Logarithmic

Lessons 2.9 through 2.15

Reviews do NOT cover all material from the lessons but will hopefully remind you of key points. To be prepared, you must study all packets from Unit 2B.

1. The function f is given by $f(x) = 3log_2(x) + 8$. Find $f^{-1}(x)$.

2. ***Calculator Active:*** What is the solution to the equation $4 \cdot 2^{x+5} - 10 = 22$.

3. What are all of the solutions to the inequality: $\ln(x + 10) + \ln 5 \leq \ln(2x + 17)$

4. ***Calculator Active:*** Mr. Sullivan used to collect football cards. When the Cleveland Browns drafted Johnny Manziel he just knew his rookie card would be worth a lot of money over time! Mr. Sullivan kept track of the value after every game Johnny Manziel played. The data is below.

Games Played	1	3	4	9
Value of Rookie Card ($)	57.50	37.77	32.74	18.55

   a) What's a logarithmic regression equation that could model this situation?

   b) How many games would Johnny Manzeil need to play for his card to be worth $10?

5. Let *a, b,* and *c* be positive constants. What is an equivalent expression to $\log\left(\frac{a^5b^2}{c^3}\right)$?

6. ***Calculator Active***: The function *f* is given by $f(x) = log_3(x)$. The function *g* is given by $g(x) = log_3(81x)$. Rewrite the function using properties of logarithms and explain any transformations from *f(x)* to *g(x)*.

7. Using the tables below, determine if the data given is exponential, logarithmic or linear.. Explain your answer.

a.

x	5	7	11	19
y	1	2	3	4

b.

x	1	2	3	4
y	1	2	4	8

c.

x	2	3	4	5
Log y	3	7	11	15

8. The function *f* is given by $f(x) = \log(3 - x) + 2$. Find the information below for *f.*

Asymptote:

Domain: Range:

End Behavior: Sketch of *f:*

9. FRQ: Directions: NO CALCULATOR
   - Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.
   - Solutions to equations must be real numbers. Determine the exact value of any expression that can be obtained without a calculator. For example, $\log_2 8$ can be evaluated without a calculator.
   - Unless otherwise specified, combine terms using algebraic methods and rules for exponents and logarithms, where applicable. For example, $2x + 3x, 5^2 \cdot 5^3, \frac{x^5}{x^2}$, and $\ln 3 + \ln 5$ should be written in equivalent forms.
   - For each part of the question, show the work that leads to your answers.

(A) The functions g and h are given by: $g(x) = \log_2(x^3) + \log_4(x)$ and $h(x) = \frac{4^{x+1}2^{3x}}{2^{2x-3}}$

i. Rewrite $g(x)$ as a single logarithm without negative exponents in any part of the expression.

ii. Rewrite $h(x)$ as an expression as a single base with an exponent.

(B) The function k is given by: $k(x) = 5 + 6\log_4(x)$. Solve $k(x) = 17$ for values of x in the domain of k.

(C) The function m is given by $m(x) = 2\ln(x) - \ln(7x - 10)$. Find all input values in the domain of m that yield an output value of 0.

10. Calculator Active

x	2	4	8	16	32
$f(x)$	8	16	24	32	40

Let f be an increasing function defined for $x > 0$. The table above gives values for $f(x)$ at selected values of x. The function g is given by $g(x) = 8.25 - 2^{x-6}$.

(A)

i. The function h is defined by $h(x) = (g \circ f)(x) = g\big(f(x)\big)$. Find the value of $h(2)$ as a decimal approximation, or indicate that it is not defined. Show the work that leads to your answer.

ii. Find the value of $f^{-1}(40)$ or indicate that it is not defined.

(B)

i. Find all values of x, as decimal approximations, for which $g(x) = 4$ or indicate there are no such values.

ii. Determine the end behavior of g as x decreases without bound. Express your answer using the mathematical notation of a limit.

(C)

i. Use the table of values of $f(x)$ to determine if f is best modeled by a linear, quadratic, exponential, or logarithmic function.

ii. Give a reason for your answer in part C (i) based on the relationship between the change in the output values of f and the change in the input values of f. Refer to the values in the table in your reasoning.

Name: _____ Date:_____ Period: _____ **Corrective**

End-of-Unit 2B Corrective Assignment –Exponential and Logarithmic

Lessons 2.9 through 2.15

Reviews do NOT cover all material from the lessons but will hopefully remind you of key points. To be prepared, you must study all packets from Unit 2B.

1. The function f is given by $f(x) = 2 \cdot 4^{x-6} - 8$. Find $f^{-1}(x)$.

2. What is the solution to the equation $5 \cdot log_2(x + 4) - 8 = 12$.

3. What are all of the solutions to the inequality: $\ln(3x - 10) > \ln(5x - 28)$

4. ***Calculator Active:*** Mr. Sullivan used collect Beanie Babies! This was a big way for him to invest money into something and hope the value would go up over time! Below he tracked the value of one of his Beanie Babies over the course of a few years.

Years Owned	2	3	5	7
Value of Beanie Baby ($)	22.12	32.56	45.85	54.60

   a) What's a logarithmic regression equation that could model this situation?

   b) How long did he own this Beanie Baby if its value is now $70?

5. Let *a, b,* and *c* be positive constants. What is an equivalent expression to $\log(3a^3b^7)$?

6. The function *f* is given by $f(x) = log_2(x)$. The function *g* is given by $g(x) = log_2(64x)$. Rewrite the function using properties of logarithms and explain any transformations from *f(x)* to *g(x).*

7. Using the tables below, determine if the data given is exponential. Explain your answer.

a.

x	5	7	11	19
y	1	2	3	4

b.

x	1	2	3	4
y	1	2	4	8

c.

x	2	3	4	5
Log y	3	7	11	15

8. The function *f* is given by $f(x) = -\log(x + 4) - 10$. Find the information below for *f.*

Asymptote:

Domain: Range:

End Behavior: Sketch of *f:*

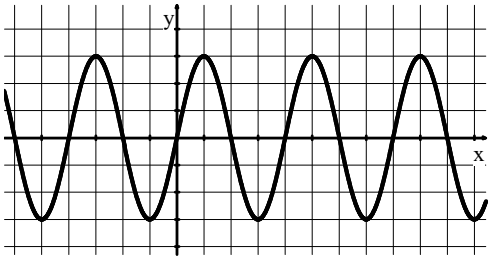
AP Precalc

Write your questions and thoughts here!

3.1 Periodic Phenomena

3.1 Notes

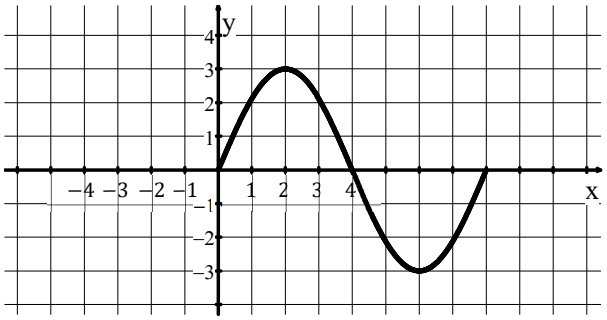
As input values increase, if the output values demonstrate a repeating pattern over successive equal-length intervals we have what is called a _____ relationship. Another word to describe this would be **cyclical**.



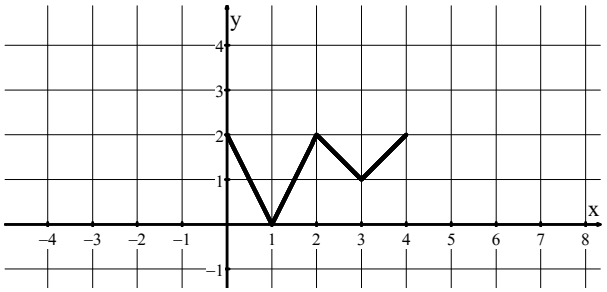
If we have the graph of a single period (cycle), then we can construct the graph of the periodic relationship. The following graphs show one period of a periodic function.

Sketch the rest of the graph on the given axes.

1.



2.

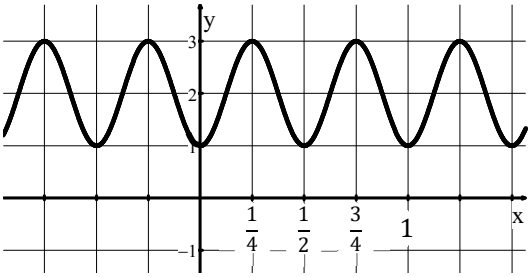


The ***period*** of the function is the smallest change in x-values it takes for the function to repeat itself. In fancy math terms, we say:

The ***period*** of the function is the smallest positive value of k such that $= f(x)$ for all x in the domain.

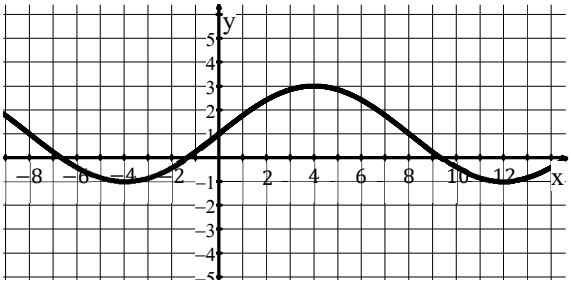
Identify the length of the period for each function.

3.



Period = _____

4.

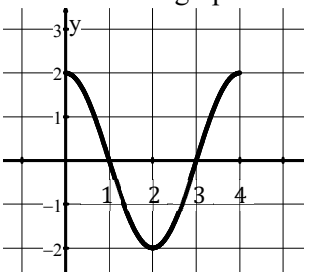


Period = _____

Write your questions and thoughts here!

Periodic functions take on various characteristics of a function such as increasing, decreasing, different concavities, etc. Recognize that any characteristics found in one period of the function will be in every period of the function.

5. Below is one cycle of a periodic function. Use the graph to answer the questions.



a. Is the function increasing, decreasing or both on the interval $18 < x < 20$?

b. Is the function concave up, concave down, or both on the interval $31 < x < 33$?

c. Is there a relative max, relative min, or neither at the point $x = 82$?

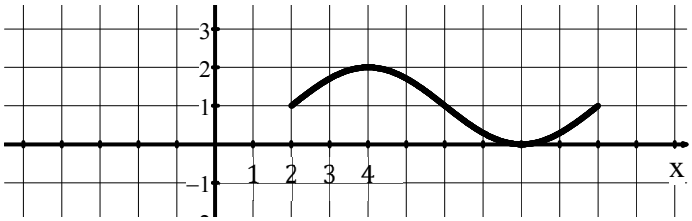
3.1 Periodic Phenomena

AP Precalculus

3.1 Practice

The following graphs show one period of a periodic function. Sketch the rest of the graph on the given axes and answer any questions that follow.

1.

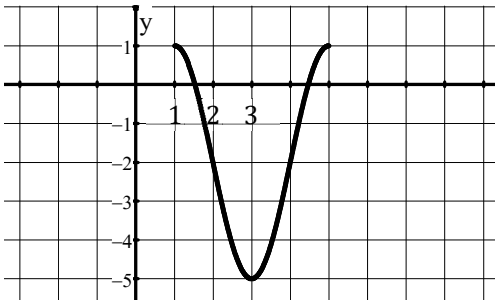


a. Is the function increasing, decreasing or both on the interval $52 < x < 56$?

b. Is the function concave up, concave down, or both on the interval $30 < x < 34$?

c. Is there a relative max, relative min, or neither at the point $x = 44$?

2.

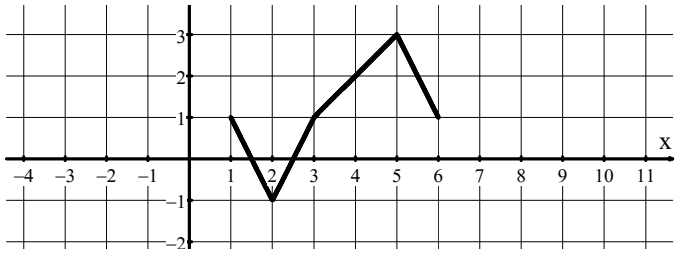


a. Is the function increasing, decreasing or both on the interval $25 < x < 27$?

b. Is the function concave up, concave down, or both on the interval $34 < x < 36$?

c. Is there a relative max, relative min, or neither at the point $x = 47$?

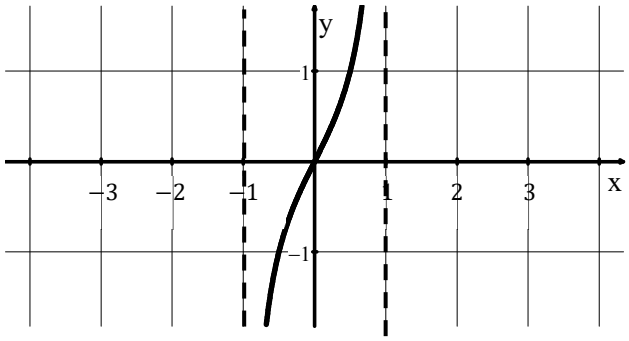
3.



a. Is the function increasing, decreasing or both on the interval $22 < x < 23$?

b. Is there a relative max, relative min, or neither at the point $x = 30$?

4.

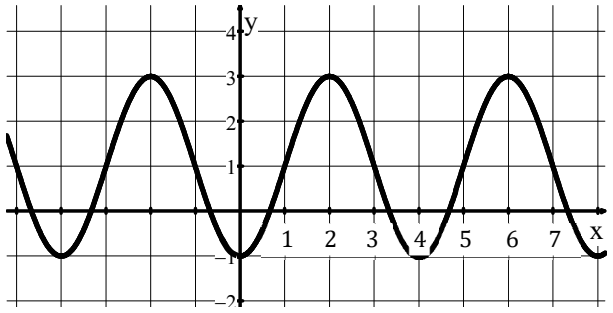


a. Is the function increasing, decreasing or both on the interval $37 < x < 39$?

b. Is the function concave up, concave down, or both on the interval $27 < x < 28$?

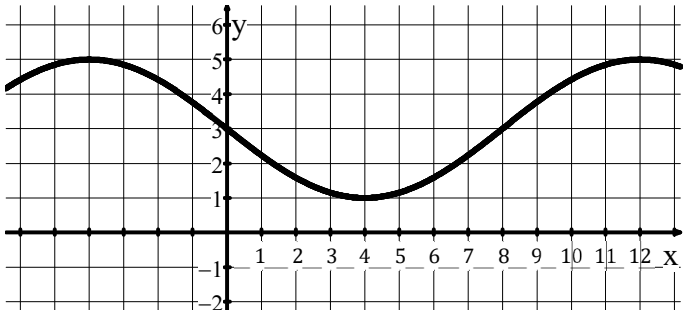
Identify the length of the period for each function.

5.



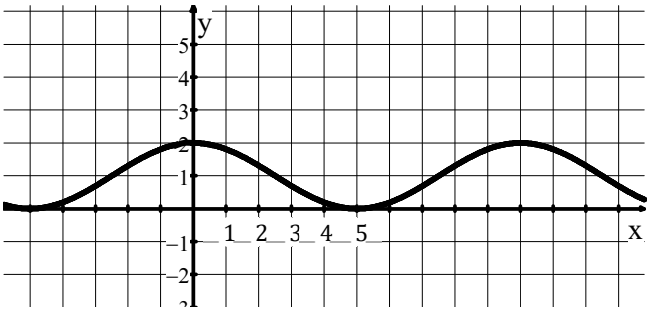
Period = _____

6.



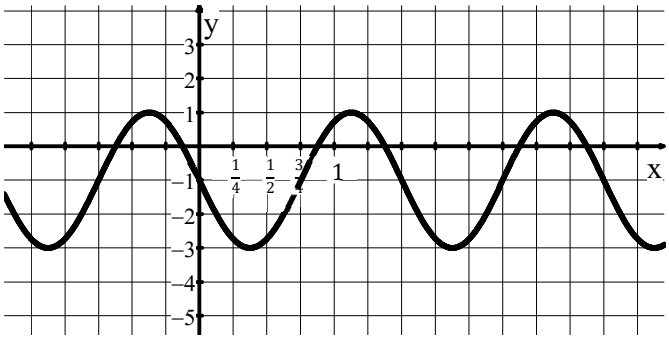
Period = _____

7.



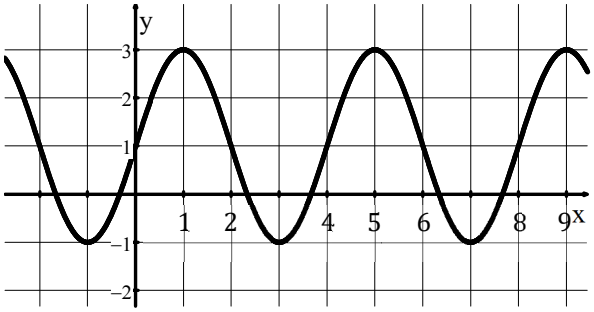
Period = _____

8.



Period = _____

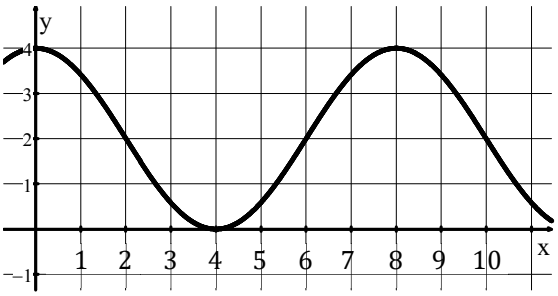
9.



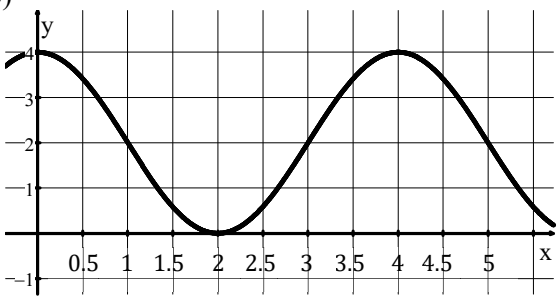
Graph of f

The graph of the function f is given in the xy-plane. Which of the following functions has the same period as f?

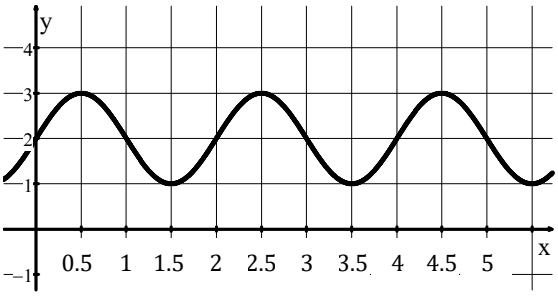
(A)



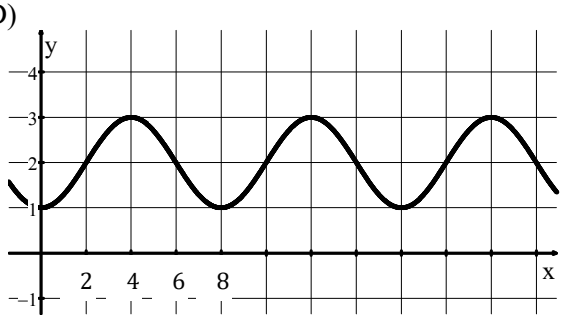
(B)



(C)



(D)

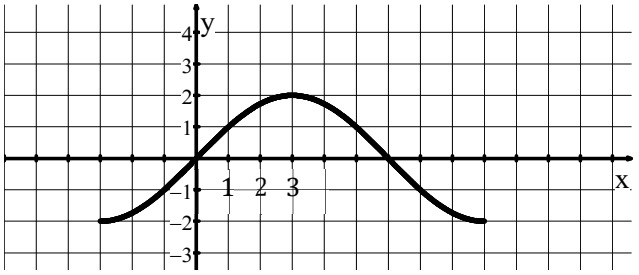


3.1 Periodic Phenomena

AP Precalculus Name: _____ **CA #1**

The following graphs show one period of a periodic function. Sketch the rest of the graph on the given axes and answer any questions that follow.

1.

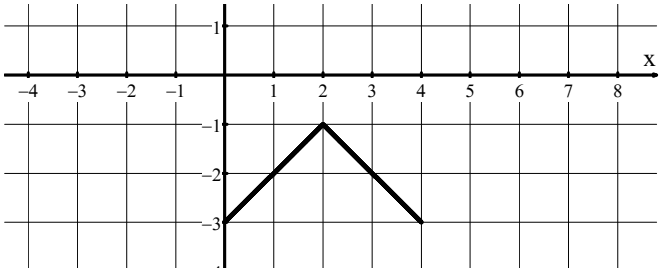


a. Is the function increasing, decreasing or both on the interval $45 < x < 51$?

b. Is the function concave up, concave down, or both on the interval $12 < x < 18$?

c. Is there a relative max, relative min, or neither at the point $x = 33$?

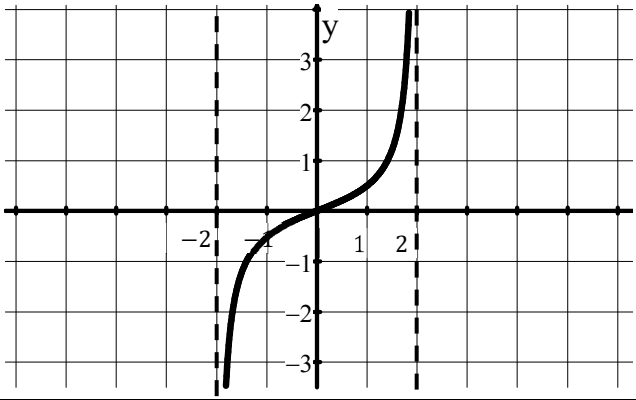
2.



a. Is the function increasing, decreasing or both on the interval $38 < x < 40$?

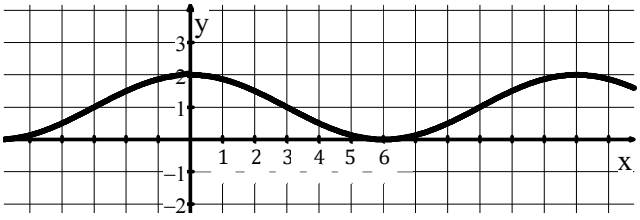
b. Is there a relative max, relative min, or neither at the point $x = 16$?

3.



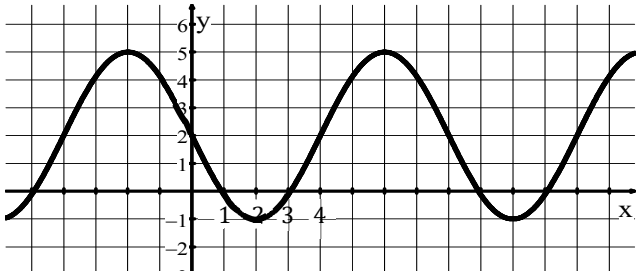
Identify the length of the period for each function.

4.



Period = _____

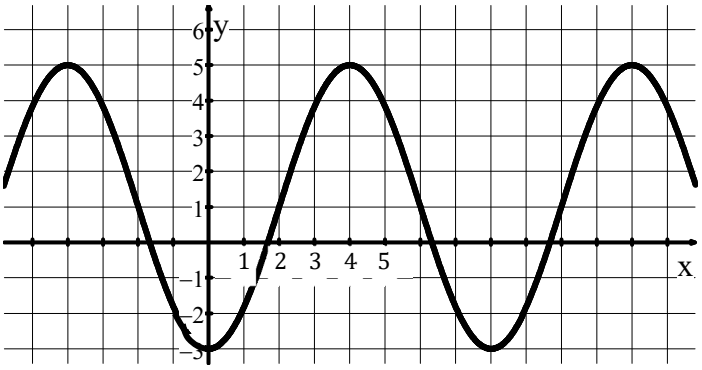
5.



Period = _____

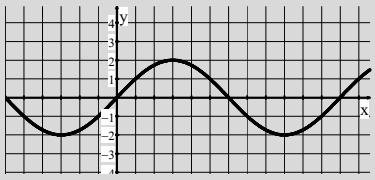
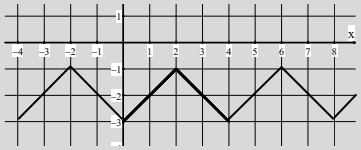
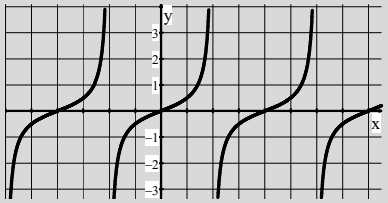
Identify the length of the period for each function.

6.



Period = _____

Answers to 3.1 CA #1

1. 	1a. Increasing 1b. Concave down 1c. Relative minimum	2. 	2a. Decreasing 2b. Relative min
3. 	4. 12	5. 8	6. 8

AP Precalc

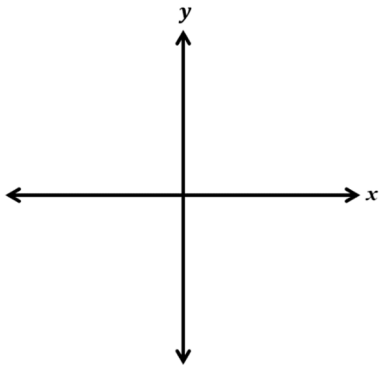
Write your questions and thoughts here!

3.2A Radians

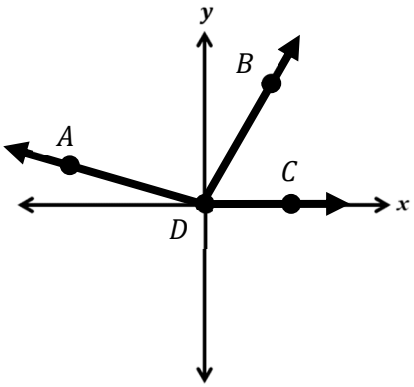
3.2A Notes

We are accustomed to finding coordinate points in a coordinate plane. But what about an angle in a coordinate plane? An angle is in *standard position* when the vertex coincides with the origin and one ray coincides with the positive x-axis. The other ray is called the *terminal ray*.

- Draw a positive 45° angle in standard position and label the initial ray and the terminal ray.
- Draw a negative 225° angle in standard position.
- Label Quadrants I, II, III, and IV.



1. Name all the angles that are in standard position. Give the initial ray and terminal ray of each angle.



Angle: _____

Initial ray: _____

Terminal ray: _____

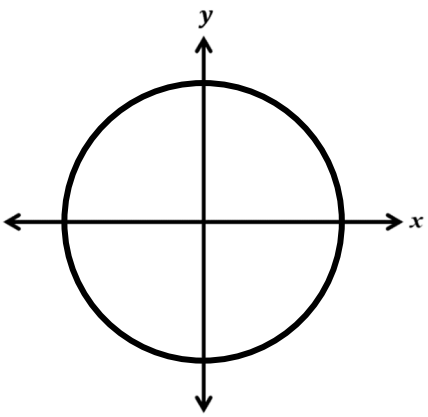
Angle: _____

Initial ray: _____

Terminal ray: _____

We are not going to measure things in degrees in this course or next year in AP Calculus. There is another way of measuring angles. That unit of measurement is called a _____. We will discuss the definition of a radian later, but first we have an important formula to learn.

Use the circle and sketch an angle with the terminal ray in QII. The rays of the angle will be touching the circle. The fancy way of saying this is that the arc of the circle is _____ by the angle. Label the circle's arc length between these rays as s. Label the radius of the circle as r. We then have



$$\boldsymbol{\theta} =$$

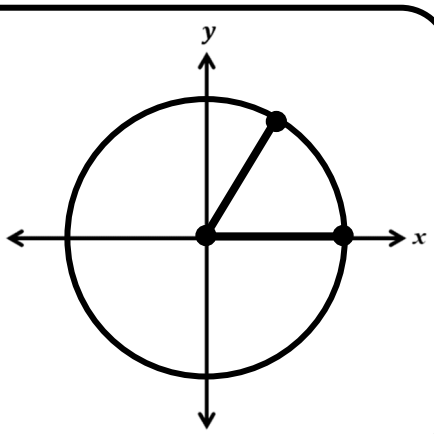
where θ is the angle measure in radians, s is the arc length subtended by rays of the angle, and r is the radius of the circle.

Write your questions and thoughts here!

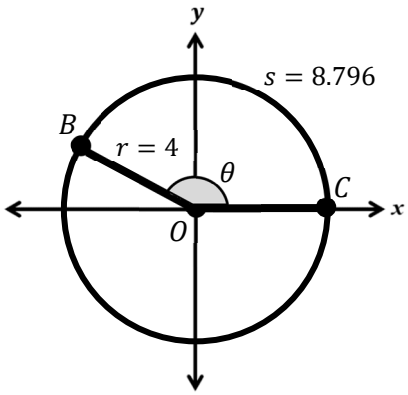
What is a Radian?

An angle of one radian is when the angle creates an arc length on a circle that is equal to the radius of the circle. This means $s = r$ and $\theta = 1$.

Can you estimate how many radians are in one circle? We will answer this later in the lesson.



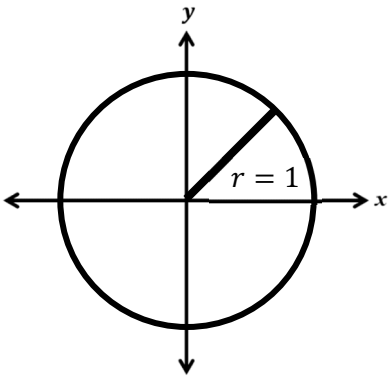
2. The figure gives an angle in standard position and a circle with radius 4 in the xy-plane. The length of the minor arc subtended by the angle is 8.796. Find the measure of the angle.



3. If an angle measure is 1.4π radians and the radius of the circle is 2.7, what is the length of the arc subtended by the angle?

4. Given an angle measure of 0.625π radians with an arc length of 3.125π, what is the radius of the circle?

Unit Circle: A circle with _____.



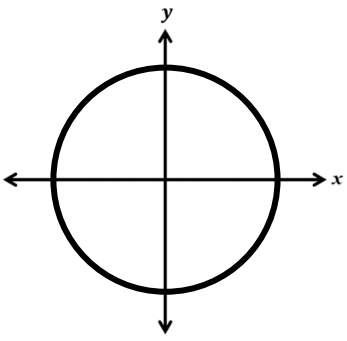
If we have a Unit Circle, then the formula for the radians simplifies.

$$\theta = s$$

The radian measure of the angle is the same as the length of the subtended arc.

For a unit circle, one full revolution would be the length of the circumference.

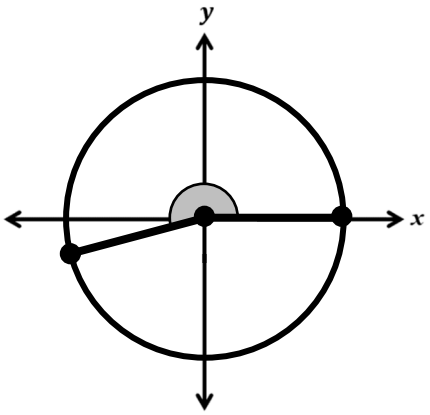
$$C = 2\pi r$$



- In other words, one revolution represents ____ radians. Was your estimate from the top of the page close?
- Half a revolution is ____ radians.
- A quarter of a revolution is _____ radians.

Write your questions and thoughts here!

5. Give an estimate of how many radians the angle represents. For simplification, leave your answer in terms of π and round the coefficient to only one or two decimal places.



Angles in standard position that share a terminal ray differ by an integer number of revolutions.

6. An angle with a measure of 4.3π would have a terminal ray in which quadrant?
Hint: Subtract or add 2π until you get a value in the interval $0 \leq \theta \leq 2\pi$.

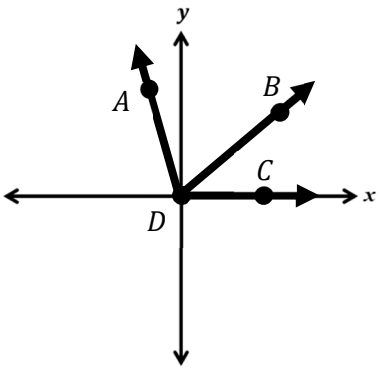
3.2A Radians

AP Precalculus

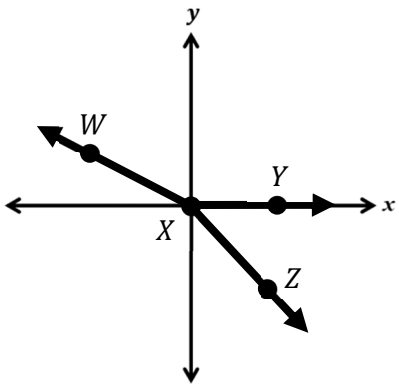
3.2A Practice

For each set of axes, name all the angles that are in standard position. Give the initial ray and terminal ray of each angle.

1.

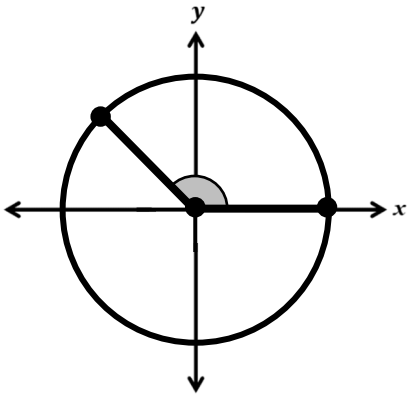


2.

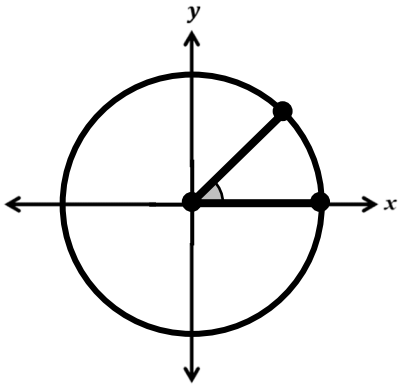


Give an estimate of how many radians the angle represents. For simplification, leave your answer in terms of π and round the coefficient to one or two decimal places.

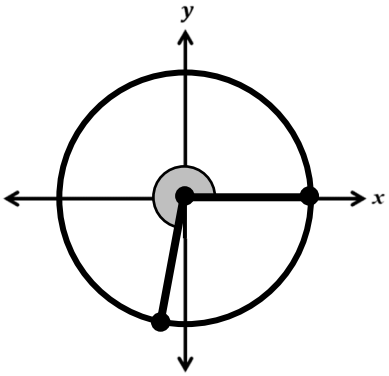
3.



4.



5.



The measurement of an angle in standard position is listed. In which quadrant is the terminal ray?

6. $\theta = 2.9\pi$

7. $\theta = 8.2\pi$

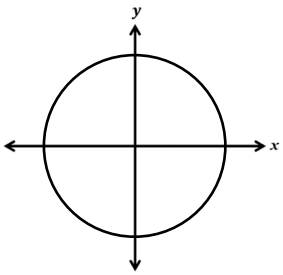
8. $\theta = 7.2\pi$

9. $\theta = -5.2\pi$

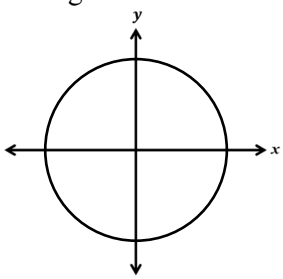
10. $\theta = -8.1\pi$

Below are various measurements of a circle's radius, an angle within the circle, or the arc subtended by the angle. SKETCH the approximate angle on the axes and find the missing value.

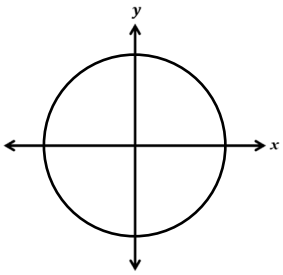
11. Radius is 2.4 and the length of an arc subtended by an angle is 1.32. Find the measure of the angle.



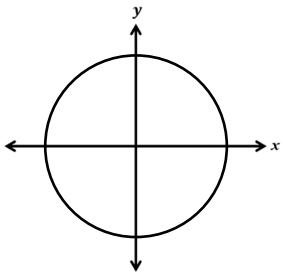
12. Radius is 1.45 and an angle is 1.4π radians. Find the length of the arc subtended by the angle.



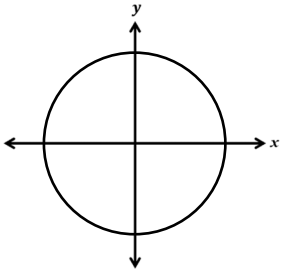
13. An angle is 0.8π radians and the length of an arc subtended by the angle is 16.9. What is the radius of the circle?



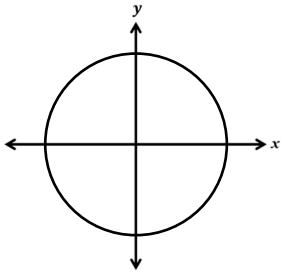
14. Radius is 6.1 and an angle is 0.5π radians. Find the length of the arc subtended by the angle.



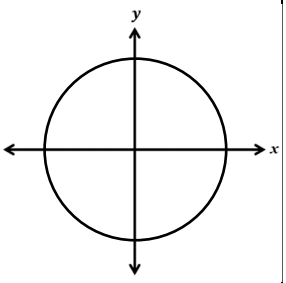
15. Radius is 6.11 and the length of an arc subtended by an angle is 30.6. Find the measure of the angle.



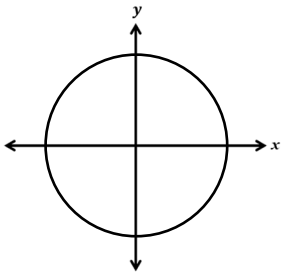
16. Radius is 6.18 and an angle is 1.7π radians. Find the length of the arc subtended by the angle.



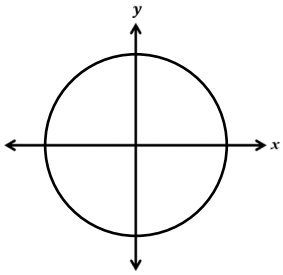
17. An angle is 4.11 radians and the length of an arc subtended by the angle is 26. What is the radius of the circle?



18. Radius is 7.8 and the length of an arc subtended by an angle is 17.94. Find the measure of the angle.



19. An angle is 3.06 radians and the length of an arc subtended by the angle is 29.47. What is the radius of the circle?



3.2A Radians

3.2A Test Prep

20. An angle with a measure of 5.7π would be equivalent to which of the following angle measures?

(A) 2.7π (B) -1.7π (C) -5.7π (D) -0.3π